

Wearables and Habits: Staggered DiD, Placebos and Shoe Bundling

Issue #7

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1 Summary

Exploratory staggered DiD (Callaway–Sant’Anna, never-treated controls, no covariates, user-quarter panel, outcome = any purchase in the quarter) of first wearable purchase on running-related and other purchases. All event studies use the universal base period, so the **reference period** is $t = -1$ (not $t = 0$); the coefficient at -1 is zero by construction.

- **Pre-trends are strong, so the reduced form is not credible as is.** Treated users purchase *less* than controls 2–8 quarters before adoption, relative to $t = -1$, in almost every outcome (Figure 1), including the placebos (Figure 2). Treated users’ purchasing is ramping up into adoption (consistent with the rising pre-adoption purchasing noted in #5).
- **No detectable post-adoption increase in running purchases.** Post ATT (quarters 1–8) for running shoes is -0.016 (0.009), running-related -0.033 (0.016), fitness gear -0.020 (0.014), supplements $+0.022$ (0.017), medication -0.005 (0.014). Negative post estimates relative to $t = -1$ mostly reflect the $t = -1$ peak in a ramp-up, not a causal drop.
- **Placebos mostly small, but the general-intensity outcomes are not.** Pet, books, gift cards, cables, cleaning, grocery and (conditional) shoe price are all within about 2 SE of zero post, yet show the same pre-trend as the main outcomes. Total purchases $\ln(1+n)$ and spend fall post (-0.11 , -0.20) after a pronounced run-up ($t = 0$ spike of $+0.20$ in $\ln(1+n)$ from the device order itself).
- **Bundling is thin.** Only 30 of the 570 adopters buy a running shoe within ± 1 month of the device (24 of 447 clean). Excluding the bundled shoe, no outcome differs significantly between bundlers and non-bundlers except fitness gear (difference -0.144 , SE 0.072, borderline), and the bundler estimates are very imprecise.
- **Running shoes as the treatment:** buying running shoes does not predict a later wearable purchase ($+0.001$, SE 0.007), but does predict later non-shoe running-related purchases ($+0.029$, SE 0.014; $+0.041$ among users who never buy a wearable), with the caveat of pre-trends. Repeat shoe purchase is $+0.082$ (mechanical replacement).

2 Design

- **Panel:** user-quarter over each user’s observed window, truncated at 2023-03 (usable window from #5); zeros filled; unbalanced panel. Built by `code/scripts/prep_design_panel.py`; estimated by `code/scripts/design_did.R`; figures/tables by `design_did_figs.py`.
- **Treated:** first non-kids wearable (Category WEARABLE_COMPUTER/BIOMETRIC_MONITOR) with ≥ 12 months pre and post (570 users); “clean” variant (447 users) also requires price $\geq \$40$,

quantity 1, not an accessory; non-clean adopters are dropped from the clean sample. **Controls:** users who never buy a wearable (about 4,100). Adopters with <12m pre/post are dropped.

- **Estimator:** `did::att_gt`, `est_method="reg"`, no covariates, universal base period, analytic SEs clustered by user, event window -8 to $+8$ quarters. “Post ATT” is the average of $e = 1, \dots, 8$ (excludes $e = 0$, which contains the device purchase itself).
- **Outcomes:** indicators for any purchase in the quarter: running shoe, running-related keyword, fitness gear, supplements, medication (definitions from #5). **Placebos:** pet, books, gift cards, cables/chargers/batteries, cleaning agents, grocery/food; $\ln$ average price of shoes (SHOES category, only quarters with a shoe purchase); and general intensity $\ln(1 + n)$ and $\ln(1 + \text{spend})$.
- **Bundling:** `bundler` = a running shoe bought within ± 1 month of the first wearable. For adopters the bundled shoe is removed from the shoe and running-related outcomes.
- **Shoe as treatment:** first running shoe (604 users with ≥ 12 m pre and post) vs. never-running-shoe users, outcomes wearable, non-shoe running-related, fitness gear, supplements, repeat shoe, with a version dropping all wearable adopters.

3 Main outcomes and placebos

Outcome	Post ATT (SE)	Pre-test p	N treated
Running shoe	-0.016 (0.009)	n/a	570
Running-related (keyword)	-0.033 (0.016)	0.000	570
Fitness gear	-0.020 (0.014)	0.000	570
Supplements	0.022 (0.017)	0.000	570
Medication	-0.005 (0.014)	0.000	570
PLACEBO: pet	-0.024 (0.016)	0.000	570
PLACEBO: books	-0.001 (0.019)	0.000	570
PLACEBO: gift cards	0.002 (0.014)	0.000	570
PLACEBO: cables/batteries	0.004 (0.019)	0.000	570
PLACEBO: cleaning	0.021 (0.014)	0.000	570
PLACEBO: grocery/food	0.020 (0.015)	0.000	570
PLACEBO: $\ln$ shoe price (cond.)	-0.046 (0.068)	0.000	441
General: $\ln(1 + \# \text{purchases})$	-0.110 (0.034)	0.000	570
General: $\ln(1 + \text{spend})$	-0.203 (0.053)	0.000	570

The pre-test column is the joint Wald test of the pre-period ATTs from `did`; it is 0.000 almost everywhere and n/a when the covariance matrix is singular, so rely on the plots. The clean-sample estimates are similar (Table in `design_tab_main_clean.tex`; running-related -0.027 (0.018), shoe -0.011 (0.011)).

4 Running-shoe bundling

Outcome	Bundlers (N=30)	Non-bundlers (N=540)	Difference
Fitness gear	-0.156 (0.071)	-0.012 (0.014)	-0.144 (0.072)
Running-related (ex-bundle)	-0.113 (0.083)	-0.016 (0.016)	-0.098 (0.084)
Running shoe (ex-bundle)	-0.005 (0.053)	-0.014 (0.009)	0.008 (0.054)
Supplements	-0.028 (0.073)	0.024 (0.017)	-0.052 (0.075)

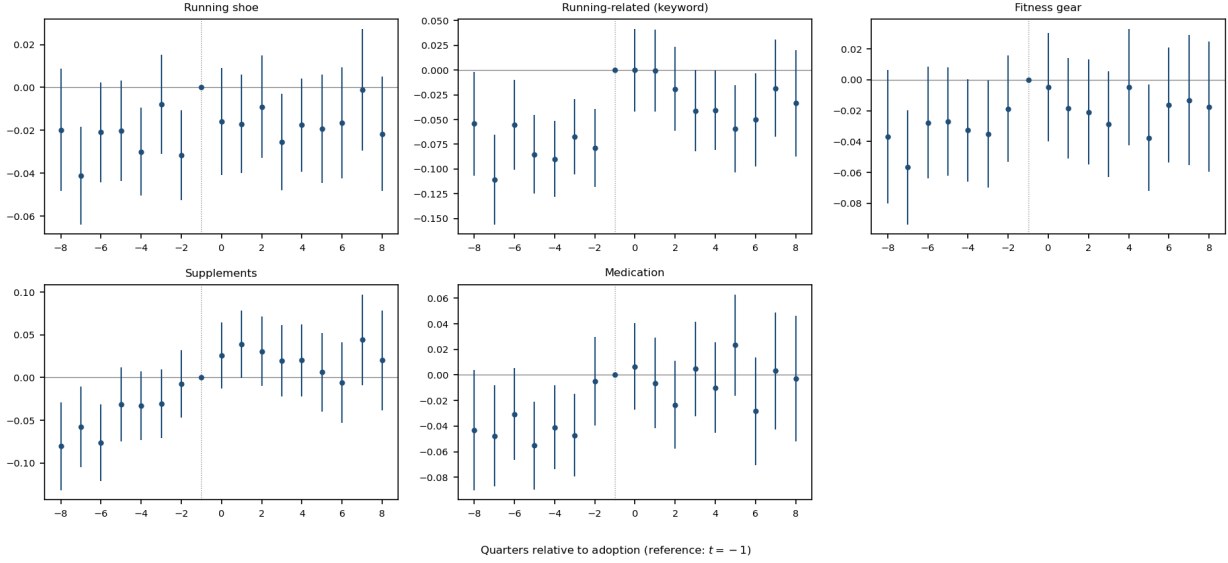


Figure 1: Event studies, main outcomes (570 adopters). 95% CIs; reference $t = -1$.

Post ATT (SE); the difference SE assumes independence (shared controls make it slightly conservative or not, unchecked). Pre-test for bundlers is not available (30 users).

5 Running shoes as the treatment

Outcome	Post ATT (SE)	Pre-test p	N treated
Wearable purchase	0.001 (0.007)	n/a	604
Running-related, non-shoe	0.029 (0.014)	n/a	604
Fitness gear	0.001 (0.013)	0.000	604
Supplements	-0.003 (0.017)	0.000	604
Running shoe (repeat)	0.082 (0.005)	n/a	604
PLACEBO-ish: medication	-0.002 (0.013)	n/a	604
PLACEBO: pet	-0.019 (0.017)	0.000	604
PLACEBO: books	-0.035 (0.018)	0.000	604

Dropping wearable adopters (399 treated): non-shoe running-related +0.041 (0.015), repeat shoe +0.074 (0.006); the wearable outcome is undefined there by construction.

6 Hypotheses for the common pre-trend (untested)

The pre-trend appears in running outcomes, supplements, medication and all placebos alike, and total purchases rise by roughly 0.35 log points from $t = -8$ to $t = -1$. That pattern points to a latent factor driving overall purchasing intensity rather than anything specific to running. Candidate explanations, with what each predicts and how it could be checked with the current data:

1. **Conditioning on activity (selection on the observation window).** Windows run from a user's first to last purchase, and adopters must have ≥ 12 months after the device. Treated users therefore are active at $t = 0$ and stay active, while the never-treated pool includes

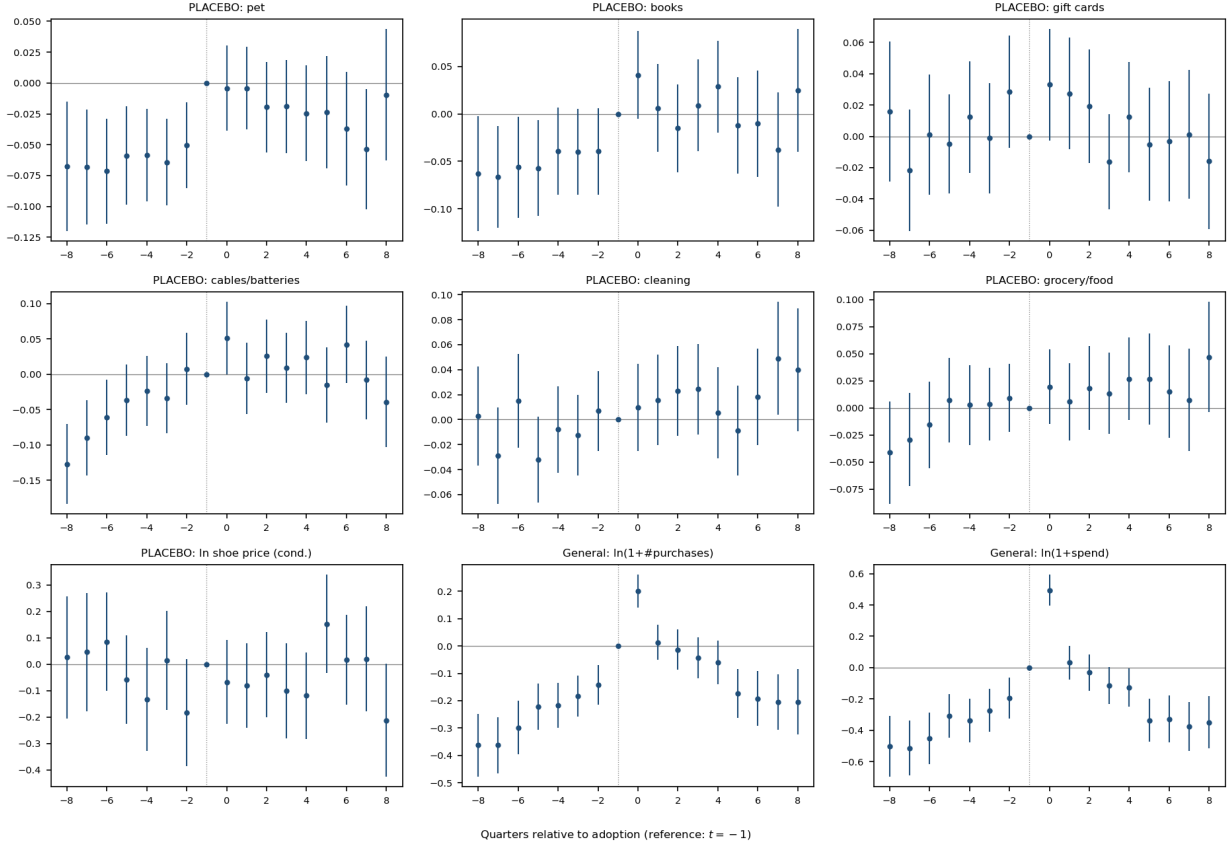


Figure 2: Event studies, placebo and general-intensity outcomes. Placebos show the same pre-trend as the main outcomes.

- dormant or short-window users. This mechanically produces a ramp-up into adoption and no matching ramp in controls. *Check:* restrict controls to users with comparable window length and activity, or match on pre-period purchase counts.
2. **Account/data maturity.** Early quarters after a user's first purchase may reflect a newly active or newly shared account, with purchasing growing over time. *Check:* plot outcomes in time since the user's first purchase for adopters and controls.
 3. **Latent health or fitness intent (anticipation).** A user who starts a fitness or health routine raises supplements, gear, medication and later the device, so purchases rise before the device. This is the plan's main threat; it would show up in health-related outcomes but should be weaker in placebos, which is not what we see, so it is unlikely to be the whole story.
 4. **Life-cycle or life-event shocks.** Moves, a new job, a new baby, a household change or New Year resolutions raise purchases across all categories and also raise the odds of buying a device. *Check:* the survey has household size and demographics; look at event studies by household type, and at adoption by quarter-of-year.
 5. **Income and tech affinity.** Adopters have higher income and larger households (#5), so they may be on a steeper growth path in Amazon spending. *Check:* covariate-adjusted or doubly-robust estimation, or income-specific trends.
 6. **Gifting and seasonality.** Devices are often bought or gifted around holidays, and holiday-heavy shoppers differ in seasonal intensity. Calendar time is handled by the control group,

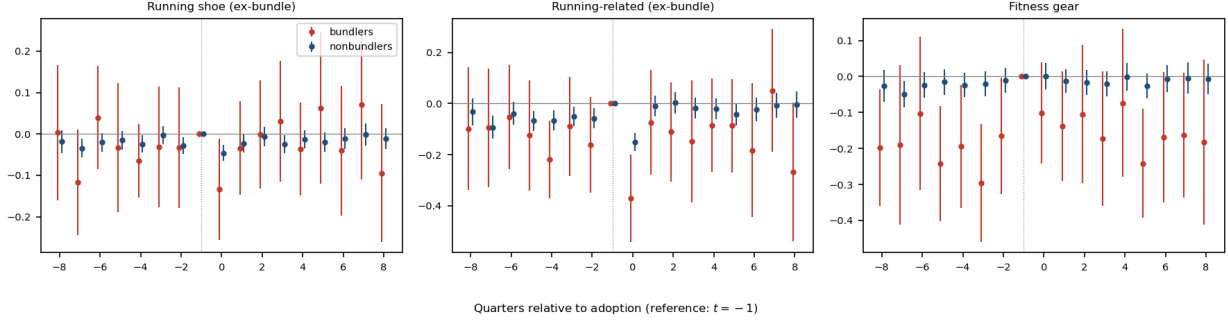


Figure 3: Bundlers (red, 30) vs non-bundlers (blue, 540), bundled shoe excluded from outcomes.

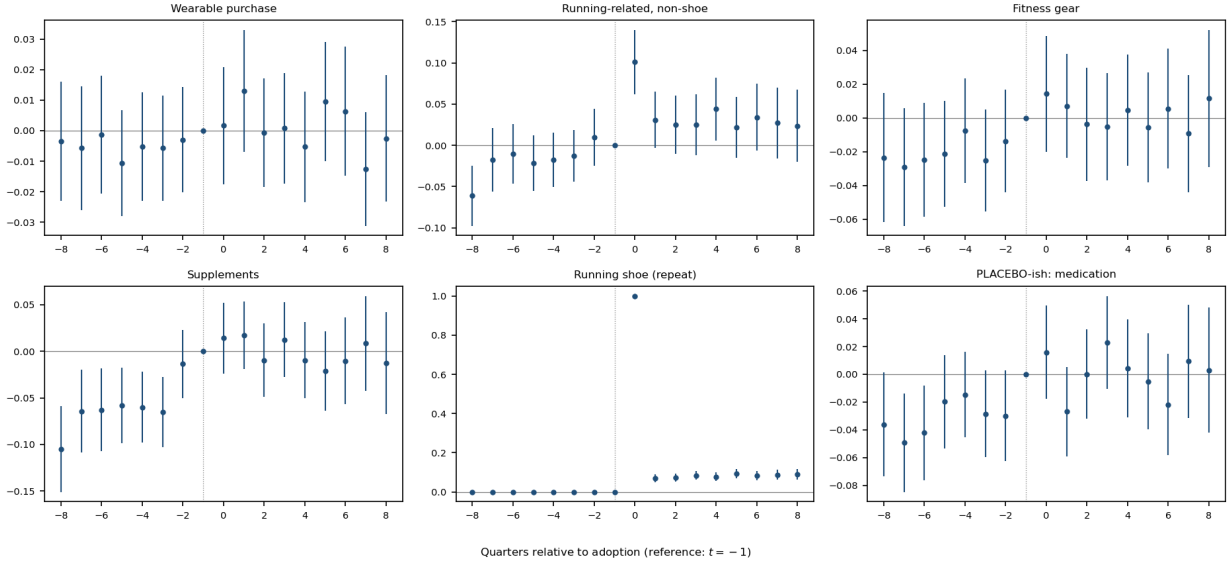


Figure 4: Event studies with the first running-shoe purchase as treatment.

but heterogeneity in seasonal intensity is not. *Check:* adoption timing by quarter and drop Q4 adopters.

7. **Measurement.** Missing Category/Title falls from 10% in 2018 to 1% in 2022, so keyword outcomes are undercounted early; this matters if adopters are observed later in calendar time than controls. The $t = 0$ spike in $\ln(1 + n)$ also reflects the device order itself.

Hypotheses 1 and 2 are mechanical and would need to be ruled out before reading any of the post-adoption estimates as behavioral. Because placebo categories move with the main outcomes, a “purchasing intensity” control (or outcomes expressed as shares of purchases) is the most direct next step.

7 Pitfalls and follow-ups

- Pre-trends (and placebo pre-trends) mean the plain unconditional DiD is not credible; follow-ups: covariates/doubly-robust, not-yet-treated controls, HonestDiD bounds, matching on pre-adoption purchasing intensity (not run here).
- Quarterly indicators with sparse purchasers and a 10% (2018) vs 1% (2022) missing Cate-

gory/Title rate make early outcomes undercounted; year effects not added.

- Bundlers are only 30 users, so the bundling comparison is underpowered; the ± 1 month window is a default.
- The conditional shoe-price placebo is estimated only on quarters with a shoe purchase (selected sample).